

Aditya Nandan

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Education

Graphic Era Deemed to be University, Dehradun
B.Tech in Computer Science Engineering

June 2027
GPA 8.54/10

Internship Experience

Quivr (YCombinator-backed) – Frontend Intern

September 2024 – November 2024

Remote

- Notetaker had no existing product surface or editor; built the feature from scratch end-to-end (8 PRs, 6 merged) using TipTap/ProseMirror, shipping a formatting toolbar, sticky search bar, and scoped styling as Notetaker 0.1
- Users needed to query their knowledge base without leaving the editor; implemented "@"-mention brain selection and a Mod+F search bar, and built a shared ChatBar component reused across search and notetaker, extending the chat/session model to support note-scoped conversations
- AI-generated text was inserted directly into notes with no way to reject it; designed a custom TipTap AIHighlight extension marking AI output as provisional, with explicit accept/decline actions before it entered the document
- The release branch had diverged 500+ files from main and editor/search-bar toggling was broken; resynced the branch and fixed the toggle interaction to unblock the 0.1 release
- Quivr had no official voice-enabled examples; built a Whisper-based STT/TTS chatbot example with translations for 11 languages, and a hands-free voice UI for the core API, both shipped with documentation

Quivr (YCombinator-backed) – Open Source Contributor

May 2023 – July 2023

Remote

- The app was a prototype with no shared UI system or Next.js routing conventions; restructured the app to the App Router pattern and built the first reusable primitives (Button, NavBar, Modal, Toast, Tooltip), establishing the component/hook/lib pattern used across the codebase
- Quivr supported only one ephemeral chat per user with no persistence; built multi-conversation chat with Supabase-backed history and a dynamic chat route, then introduced a React context layer after discovering the initial hook caused duplicate API calls and state across components
- Knowledge bases were single and implicit, blocking per-use-case organization; designed a multi-brain architecture and shipped a searchable brain switcher on Radix UI popovers with inline creation and deletion, after iterating past an unmerged localStorage-based prototype
- Recurring hydration errors, render loops, and layout breakage were destabilizing the app; traced a hydration bug to a server/client env-variable mismatch and fixed it at the source, resolved a chat page infinite-render loop, and shipped responsive layouts for navbar, upload, and explore

Projects

Heimdall: Local AI Screen Memory for Hyprland (Wayland)

github.com/adityanandanx/heimdall/

- Event-driven Python capture daemon on the Hyprland socket: on window changes, captures the active-window region (grim) and its accessibility tree (AT-SPI) into a searchable SQLite/FTS5 store; falls back to RapidOCR on the Intel NPU (OpenVINO) with CPU fallback for blind windows
- YouTube watch-sessions (MPRIS + Chromium MV3 extension + yt-dlp/faster-whisper transcripts) become searchable via subtitles; FastAPI serving search / frames / sessions / a live settings interface
- "Day recap" and "time breakdown" pipes (APScheduler) run through a local Gemma 4 QAT model on the Arc Vulkan backend (llama-server); Tauri 2 desktop client for day timeline, search, sessions, status and settings. Everything stays on-machine
- Built with the help of opencode following a ticket based task management AI-native workflow - decomposing features into scoped tickets and driving agent-executed implementation

Hybrid RAG pipeline: Conversational Movie Agent

github.com/adityanandanx/cinema-rag

- Built a multi-agent hybrid Retrieval Augmented Generation (RAG) pipeline that answers natural language queries across 2200 movie screenplays chunked into 289124 documents using dense and sparse retrieval
- Used the BAAI/bge-small-en-v1.5 embedding model for dense vector embeddings, BM25 for sparse keyword retrieval and Reciprocal Ranked Fusion (RRF) to combine the results.

- Used Qdrant running in a local Docker container as the vector database and Llama 4 17b as LLM for response generation via Groq cloud for fast inference.

BitFlip: Story Based 2D Platformer Game

github.com/Team-Hacksmith/bitflip

- Created a 2D platformer in 72 hours, winning 1st place at IIT Ropar's Advitiya Game Jam 2025
- Constructed with Godot using GDScript; led a 3-member team and optimized for smooth performance
- Implemented advanced mechanics such as coyote time, variable jump height, and jump buffering

Nirbhay: Automated Emergency Detection App

github.com/Team-Hacksmith/nirbhay

- Built an accessibility-focused safety app that automatically detects distress and triggers hands-free emergency alerts
- Integrated wearable hardware (ESP32), a Flutter mobile app, and a Firebase backend with YAMNet for real-time audio distress detection
- Empowered elderly, disabled individuals, and at-risk populations to summon help without manual intervention

Skills

- **Languages:** Python, TypeScript, JavaScript, Java, C, C++, PHP
- **AI/ML:** LangChain, LangGraph, Hybrid RAG, Agentic AI, Generative AI, Computer Vision, NLP
- **Frontend:** React.js, Next.js, Vue.js, Flutter, Tailwind CSS, SCSS
- **Backend:** FastAPI, Flask, Django, Laravel, Node.js, Express.js
- **Databases:** PostgreSQL, MySQL, MongoDB, Firebase
- **Cloud & DevOps:** AWS (S3, EC2, Lambda), Git, Docker

Honors and Awards

WorldSkills Asia: Web Technologies

Nov 2025

WorldSkills Asia Taipei, Taiwan

- Represented India in Web Technologies against 14 countries across 5 independent 3-hour modules (speed test, frontend, backend, fullstack, team app) over 3 days, each with a fresh spec and zero internet access; only offline documentation provided
- Built each module from scratch to a pixel-accurate design and marking-scheme test cases within the 3-hour window using Laravel, Next.js, React, Vue, and Nuxt, relying solely on offline docs with no external references or libraries
- Won Medallion of Excellence, ranked 5th globally

IndiaSkills National Qualifier: Web Technologies

2024

Central Government of India, Ministry of Skill Development & Entrepreneurship

- Needed to place top nationally to qualify for WorldSkills Asia; underwent 2+ months of intensive training and 3 internal test rounds simulating competition constraints (fixed time, blind specs, no external resources)
- Won IndiaSkills National 2024 in Web Technologies, securing selection to represent India internationally

SWOC 2024 Winner

March 2024

Social Winter Of Code

- Recognized as the #1 Open Source contributor out of 20,000+ developers
- Contributed to 10+ projects and resolving 80+ issues to open-source repositories

IIT Ropar Advitiya Game Jam 1st Prize

2025

IIT Ropar

- 1st place, BitFlip, nation-wide 72-hour game development hackathon

HackTheHills Hackathon 2nd Prize

2024

DIT University

- 2nd place, ScripturaAI, national coding hackathon with 50+ teams

AWS Jam Grandmaster

2025

Graphic Era University

- Winner, campus-wide cloud and DevOps hackathon for Amazon Web Services challenges